```
(%i3) annuity_fv(0.08, 100000, 5);
(%o3)                    17045.64545668365
(%i4) quit();
```

Project planning

Finance management is a key element of planning any project, whether professional or personal. Assume a project would take 'n' years, with given yearly expenses, and say the available interest rate is 'r' p.a. Then, a common question that every project manager needs to answer is: what is the net present value (NPV) of the project, which needs to be invested for the project. The answer to this basic question is, more often than not, one of the important factors for deciding whether or not to take up this project. Maxima provides *npv()* to compute the same. As an example, if a project needs ₹100, ₹200, ₹150 and ₹600 over four years, respectively, and the current interest rate is 7 per cent, what is the NPV? It would be ₹848 as shown below:

```
$ maxima -q
(%i1) load(finance)$
(%i2) npv(0.07, [100, 200, 150, 600]);
(%o2)                    848.3274983420189
(%i3) quit();
```

Another common practice when choosing between various projects is to compute the benefit-to-cost ratio of the various projects, and select the one with the best ratio. The benefit-to-cost ratio for a given interest rate <*r*> could be computed using *benefit_cost()*. Here's an example to demonstrate this, assuming 18 per cent as the rate of interest:

Project P1 (2 years): Yearly benefits (100, 200), yearly costs (150, 50)

Project P2 (3 years): Yearly benefits (0, 200, 100), yearly costs (100, 100, 0)

Project P3 (4 years): Yearly benefits (0, 200, 200, 100), yearly costs (100, 100, 50, 50)

```
$ maxima -q
(%i1) load(finance)$
(%i2) benefit_cost(0.18, [100, 200], [150, 50]);
(%o2)                    1.400881057268722
(%i3) benefit_cost(0.18, [0, 200, 100], [100, 100, 0]);
(%o3)                    1.306173223448919
(%i4) benefit_cost(0.18, [0, 200, 200, 100], [100, 100, 50, 50]);
(%o4)                    1.489492494361802
(%i5) quit();
```

And clearly, over the long run, the four-year project P3 has a better benefit-cost ratio. But if one is only looking for shorter term benefits, then you could go with project P1 as well. END

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